NIRBHAYANET

PROTECT. CONNECT. RECOGNIZE.

A Community-Powered Women's Safety Ecosystem

A Project Report submitted for the SEVA FIRST Innovation Challenge 2026

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INTRODUCTION

Background :

Women's safety is an important social concern in both urban and rural environments. Women may encounter safety-related difficulties while commuting, travelling alone, or visiting unfamiliar locations. In such situations, access to reliable safety information, trusted contacts, emergency resources, and appropriate local assistance can be valuable.

Digital technologies provide opportunities to improve access to safety-related information and communication. Smartphones, location services, web applications, and notification systems can support features such as journey sharing, access to emergency resources, and the discovery of participating safe places.

However, these capabilities may be distributed across different applications and services. A coordinated platform could make relevant safety features easier to access through a common interface.

Project Overview:

NIRBHAYANET is a proposed community-powered women's safety ecosystem designed to coordinate multiple safety-related functions through a digital platform.

The platform focuses on emergency support access, journey sharing, trusted contacts, community assistance, participating safe places, public infrastructure reporting, and responsible community contribution recognition.

The system is designed around a modular approach so that individual features can be developed, integrated, and tested progressively.

Motivation:

The motivation behind NIRBHAYANET is to explore how digital technology and responsible community participation can support women's safety.

The project emphasises convenient access to safety features, user control over information sharing, appropriate verification of participating volunteers, and clear distinctions between platform-based assistance and official emergency services.

Scope of the Project:

The project scope includes:

- Providing access to emergency helpline information.
- Supporting journey sharing with selected trusted contacts.
- Exploring location-based discovery of participating safe places.
- Supporting structured community assistance workflows.
- Providing a mechanism for submitting public safety concerns.
- Supporting authorised administration and partner management.
- Considering privacy, security, and user consent throughout development.

The availability of each function depends on its implementation and testing status. Features that remain incomplete are treated as planned functionality rather than completed capabilities.

Intended Users:

The intended users and stakeholders include women students, working women, women travelling alone, eligible community volunteers, participating local businesses, partner organisations, and authorised administrators.

ABSTRACT

Women's safety remains an important concern, particularly during daily commuting, travelling alone, and accessing unfamiliar public spaces. Although emergency helplines, navigation platforms, and personal safety applications provide useful services, users may still face challenges in coordinating trusted contacts, finding participating safe places, and accessing appropriate community assistance.

NIRBHAYANET is a technology-driven women's safety platform designed to bring these safety-related functions together within a coordinated digital ecosystem. Its proposed features include emergency support access, journey sharing, trusted contacts, a nearby help network, a safe places directory, community safety reporting, and recognition of responsible community contributions.

The platform follows a modular software architecture intended to support a user-friendly interface, backend processing, data management, and integration with relevant external services. Privacy, access control, responsible volunteer participation, and user consent are important considerations in the proposed design.

The project aims to improve access to safety information and facilitate coordination among women, trusted contacts, participating community members, and partner organisations. NIRBHAYANET is intended to complement existing emergency services rather than replace them.

The current project includes implemented and pending components. Its functionality and readiness must therefore be evaluated according to the features actually developed and the results of documented testing.

PROBLEM STATEMENT

Women may face safety concerns while travelling, commuting, or accessing unfamiliar public spaces. During such situations, finding appropriate assistance, sharing journey information with trusted contacts, and locating reliable nearby support may be challenging.

Existing emergency services, navigation platforms, and personal safety applications provide important capabilities. However, coordinating different safety-related needs through a single accessible platform remains an area that can be explored.

The problem addressed by NIRBHAYANET is the need for a coordinated, privacy-conscious digital platform that brings together access to emergency resources, journey sharing, trusted contacts, participating safe places, and structured community assistance.

The platform is intended to complement established emergency services and must not create the impression that community volunteers or digital features can guarantee personal safety.

OBJECTIVES OF THE PROJECT

The main objective of NIRBHAYANET is to develop a coordinated digital platform that supports women's safety through accessible technology and responsible community participation.

The specific objectives are:

- To provide convenient access to emergency helpline information.
- To support journey sharing with user-selected trusted contacts.
- To facilitate the discovery of participating safe places.
- To design a structured workflow for community assistance, subject to volunteer eligibility and availability.
- To provide a mechanism for reporting public infrastructure and safety concerns.
- To explore responsible recognition of community contributions.
- To provide appropriate administrative functions for managing participating partners and submitted reports.
- To consider user consent, access control, and privacy-conscious data handling.
- To develop and evaluate the system incrementally through functional testing.
- To create a modular foundation that can be improved through future development and user feedback.

EXISTING SYSTEM AND ITS LIMITATIONS

Existing System:

Several digital platforms and public services provide features that support personal safety, emergency response, navigation, and location sharing. These services play an important role in helping users access information and assistance.

Examples include government emergency response services, personal safety applications, digital mapping platforms, and informal community support networks.

Emergency Response Services:

Government emergency response services provide access to emergency assistance. In India, the Emergency Response Support System uses the helpline number 112.

These services are important resources during emergencies. NIRBHAYANET is designed to complement such services rather than replace them.

Personal Safety Applications:

Personal safety applications may provide features such as SOS alerts, location sharing, and emergency notifications. The availability of these features varies by application.

Digital Mapping Platforms:

Digital mapping platforms provide navigation, location search, and geographical information. They can help users identify places and plan journeys.

Community Support Networks:

Community organisations, local businesses, and individuals may offer assistance to people who need help. However, the availability and organisation of such support vary by location and participating organisation.

Limitations and Opportunities:

The following points identify opportunities for a coordinated platform. They should not be interpreted as limitations of every existing service.

- **Distributed Features:** Users may need to access different services for emergency information, navigation, journey sharing, and other safety-related needs.
- **Coordination Challenges:** Coordinating trusted contacts and community assistance may require multiple communication channels.
- **Safe-Place Information:** Information about participating safe places may not be available through a single dedicated directory in every location.
- **Community Assistance:** Community assistance requires suitable eligibility checks, availability information, accountability, and safeguards.
- **Privacy Considerations:** Location sharing and personal information require appropriate consent and access controls.
- **Service Availability:** Digital safety features depend on factors such as internet connectivity, device availability, and the availability of external services.

Need for an Improved Approach:

These observations motivate the development of NIRBHAYANET as a coordinated platform for selected women's safety workflows.

The proposed platform aims to make relevant features accessible through a common interface while maintaining appropriate privacy safeguards and complementing existing emergency services.

PROPOSED SYSTEM

Overview:

NIRBHAYANET is a proposed digital women's safety platform designed to bring together safety information, trusted-contact communication, journey sharing, participating safe places, community assistance, and public safety reporting.

The system follows a modular architecture in which individual features can be developed and tested independently before integration.

The platform aims to improve access to relevant safety resources and support coordination among users, trusted contacts, eligible community volunteers, participating organisations, and authorised administrators.

Main Features of the Proposed System:

- **Emergency SOS and Helpline Access:**

This module is intended to provide convenient access to emergency helpline information and support the application's implemented emergency workflow.

Any direct integration with emergency services must be separately implemented, authorised where necessary, and tested.

- **Smart Journey Guardian:**

This feature is designed to let users share relevant journey information and location updates with selected trusted contacts, subject to permissions and the availability of location services.

- **Nearby Help Network:**

This module proposes a structured process for requesting assistance from eligible community volunteers when available. Volunteer participation must follow defined verification and safety procedures.

- **Trusted Contacts:**

Users can identify trusted people with whom they may share permitted safety updates. Information sharing should remain subject to user consent and appropriate access controls.

- **Safe Places Network:**

This module is intended to help users discover participating safe places, such as verified partner establishments or community locations.

Listings should be checked and maintained through a defined verification process. Participation does not guarantee safety in every situation.

- **Community Safety Map:**

The proposed map feature can present relevant safety information and reports concerning public infrastructure, subject to data availability and appropriate review.

- **Good Samaritan Recognition:**

This feature proposes recognising responsible community contributions through a transparent recognition mechanism. Recognition should be based on defined criteria and should not encourage unsafe intervention or unverified claims.

- **Administration and Partner Dashboard:**

The administrative module is intended to help authorised personnel manage participating partners, review submitted reports, and oversee volunteer eligibility according to their assigned permissions.

Advantages of the Proposed System:

The proposed approach offers the following potential advantages:

- Multiple safety-related functions can be accessed through one platform.
- Journey information can be shared with selected trusted contacts.
- Participating safe places can be organised in a dedicated directory.
- Community assistance can follow a structured workflow.
- Public safety concerns can be submitted for review.
- Modular development allows features to be tested and improved incrementally.
- Privacy and access control can be incorporated into the system design.

Safety and Privacy Considerations:

NIRBHAYANET should follow privacy-conscious design principles, including collecting only necessary information, restricting access to authorised users, and obtaining appropriate consent before sharing location or journey information.

The platform must not imply that volunteer availability, a safe-place listing, or a digital notification guarantees personal safety. Existing emergency services must remain independently accessible.

Expected Outcome:

The expected outcome is a software platform that provides a coordinated interface for selected women's safety workflows. The final capabilities will depend on the features implemented, the external services available, and the results of functional, usability, and security testing.

SYSTEM REQUIREMENTS

Introduction:

System requirements define the hardware, software, and supporting resources needed to develop, run, and test NIRBHAYANET. The requirements depend on the technologies selected and the features implemented in the project.

Hardware Requirements:

The following hardware may be used for development and testing:

- **Computer/Laptop:** A computer with a dual-core processor or equivalent for application development and execution.
- **RAM:** A minimum of 4 GB RAM to run the development environment and related tools.
- **Storage:** At least 10 GB of available storage space for source code, software, project files and related resources.
- **Display:** A standard monitor or laptop display for coding and interface design.
- **Internet Connection:** A stable internet connection for research, team collaboration and accessing online services.
- **Smartphone:** An Android or iOS device for testing the responsive interface and mobile usability.
- **Input Devices:** A standard keyboard and mouse or touchpad for development and system interaction.

Software Requirements:

NIRBHAYANET requires suitable software tools and development technologies to build, manage, and test its application. The following software components may be used depending on the project's actual implementation.

- **Operating System:** Windows, Linux, or macOS for application development and testing.
- **Code Editor:** Visual Studio Code for writing and editing source code.
- **Frontend Technologies:** HTML, CSS, JavaScript, and React.js for developing the user interface.
- **Backend Technology:**

1. **Supabase:** Backend as a Service (BaaS).
2. **PostgreSQL:** Database management for data storage.
3. **SQL:** Database tables, queries and Row Level Security (RLS) for policies.
4. **Supabase JavaScript Client ("@supabase/supabase-js"**): React is used to connect supabase backend with frontend.
5. **JavaScript/TypeScript:** Frontend and browser-side for application login.

- **Web Browser:** Google Chrome, Microsoft Edge, or Mozilla Firefox for accessing and testing the web application.
- **Version Control:** Git and GitHub for source-code management and team collaboration.
- **External Services:** Mapping, geolocation, and notification services, where required and implemented.

External Services:

Depending on the implemented features, the project may require:

- Mapping and geolocation services.
- Notification services.
- Access to relevant emergency helpline information.
- Hosting or deployment services.
- External services may have usage limits, configuration requirements, or additional costs.

Development Environment:

The application can be developed using standard computing resources and software development tools. No specialised fabrication equipment is required for the software prototype.

Conclusion:

NIRBHAYANET is a software-based project that can be developed and tested using commonly available computing resources. The final requirements depend on the selected technology stack and the application's implementation status.

System Architecture

Explanation:

Users interact with the React frontend. Application modules manage safety-related actions. Supabase connects the application to the PostgreSQL database, while Row Level Security (RLS) restricts access to authorised records.

- **Users Layer:**

USERS: The top-level entry point representing all system participants, branching into two specific user groups:

Women Users: The primary end-users accessing the core safety features.

Administrators: Privileged users responsible for management and oversight.

- **Presentation Layer:**

REACT FRONTEND: Acts as the centralized user interface layer where requests from both Women Users and Administrators converge and are processed.

- **Application Modules Layer:**

SAFETY APPLICATION MODULES: The core functional layer handling safety logic, which splits into three specific modules:

SOS: Emergency alert and response handling.

Trusted Contacts: Management of emergency contacts and user connections.

Journey Safety: Real-time tracking and safety monitoring for commutes.

- **Integration & Middleware Layer:**

SUPABASE CLIENT/API: Manages communication, API requests, and data synchronization between the frontend modules and the backend database.

SECURITY AND RLS LAYER: Implements security protocols and Row-Level Security (RLS) policies to ensure authorized access and data privacy.

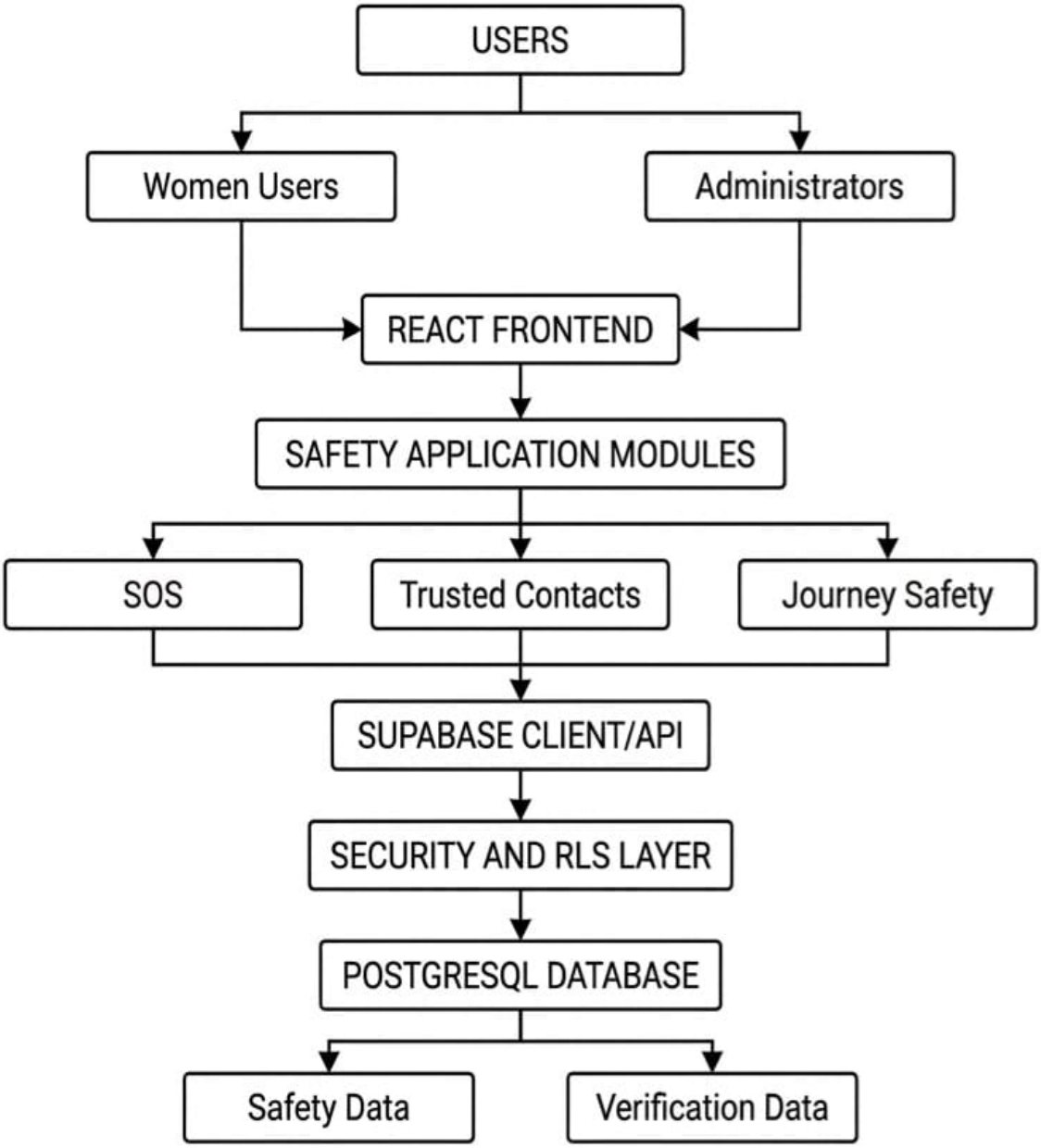
- **Data Layer:**

POSTGRESQL DATABASE: The core relational database management system storing the application's persistent data, which splits into two distinct categories:

Safety Data: Stores logs, alerts, journey records, and emergency contact details.

Verification Data: Stores user authentication credentials and administrative verification logs.

Architechture:



WORKING PRINCIPLE AND METHODOLOGY

Introduction:

This chapter describes the proposed working principle and development methodology of NIRBHAYANET. The platform is designed to provide a coordinated interface for selected women's safety functions, including emergency information, journey sharing, trusted contacts, participating safe places, and community assistance.

The workflow of each feature depends on its implementation and the external services available.

Working Principle:

The proposed system follows a sequence of user interaction, request processing, data management, and response generation.

- **Step 1: User Access**

The user accesses the application through its available interface. If registration and authentication have been implemented, the user can access the features permitted for their account.

- **Step 2: Feature Selection**

The user selects an available feature, such as journey sharing, viewing participating safe places, or accessing emergency helpline information.

- **Step 3: Request Processing**

The frontend sends the required request to the backend, where applicable. The backend validates the request and checks relevant permissions.

- **Step 4: Data Processing**

The application retrieves or updates the necessary information in the database. If the selected feature requires an external service, the application communicates with that service when integration is available.

- **Step 5: Response and Status Display**

The application returns the result or current status to the user interface. Relevant information may be displayed to the user or shared with authorised contacts according to the feature's implementation and permissions.

- **Step 6: Feedback and Reporting**

Where implemented, users can submit feedback or safety reports for appropriate review.

Development Methodology:

NIRBHAYANET follows an incremental development approach. The application is divided into manageable components that can be developed and tested progressively.

The methodology includes the following stages:

- **Requirement Analysis:** Identify the problem, intended users, and proposed features.
- **System Design:** Plan the architecture, user interface, database, and workflows.
- **Frontend Development:** Develop the user interface and navigation.
- **Backend Development:** Implement application logic and required APIs.
- **Database Development:** Design the data structure and relationships.
- **Integration:** Connect implemented frontend, backend, database, and external services.
- **Testing:** Check functionality, input validation, access permissions, and error handling.
- **Improvement:** Resolve identified issues and refine the application.

Safety Considerations:

The system should preserve user control over information sharing and restrict sensitive information to authorised users. Volunteer assistance should depend on defined eligibility checks and actual availability.

Emergency helpline access must remain independent of volunteer availability. The application should not imply that a digital notification or community response guarantees safety.

Conclusion:

The proposed methodology supports gradual development and evaluation of NIRBHAYANET. Each feature must be tested according to its actual implementation status before being presented as operational.

MODULE DESCRIPTION

Introduction:

NIRBHAYANET is organised into functional modules. Each module addresses a particular aspect of women's safety and community participation. The modules can be developed and integrated progressively.

Emergency SOS and Helpline Access:

This module is intended to provide convenient access to emergency helpline information and support the application's implemented emergency workflow.

Any direct connection to external emergency services must be implemented and tested separately. The platform is not a replacement for official emergency response services.

Smart Journey Guardian:

This module is designed to support journey sharing with selected trusted contacts.

Depending on the implementation, it may display journey details and provide location updates when the necessary permissions and location services are available.

Nearby Help Network:

This module proposes a structured process for requesting assistance from eligible community volunteers.

Volunteer participation should depend on defined verification procedures, availability, and appropriate safeguards. The system must not guarantee that a volunteer will be available.

Trusted Contacts:

This module allows users to identify trusted people with whom they may share permitted safety-related information.

Access to journey information and notifications should follow user preferences, consent, and applicable permissions.

Safe Places Network:

This module is intended to display participating safe places, such as verified partner establishments or community locations.

The accuracy of listings depends on verification, maintenance, and available location information. A listing must not be treated as a guarantee of safety.

Community Safety Map:

This module proposes a map-based view of relevant safety information and submitted public infrastructure concerns.

Reports should be reviewed appropriately, and unverified submissions should not be presented as confirmed facts.

Good Samaritan Recognition:

This module proposes a mechanism for recognising responsible community contributions.

Recognition criteria should be transparent and should not encourage risky intervention, inappropriate competition, or unverified claims.

Administration and Partner Dashboard:

This module is intended for authorised administrators and participating partners.

Depending on the implemented functionality, it may support partner management, volunteer eligibility review, and the handling of submitted safety reports. Access should be restricted according to assigned roles.

Module Integration:

The modules are intended to work within a common application architecture. The frontend provides the interface, the backend processes requests, and the database stores relevant information. External services may support selected functions where integration is available.

Conclusion:

The modular structure provides a foundation for developing and maintaining NIRBHAYANET. The functionality of each module must be documented according to what has actually been implemented and tested.

DATABASE DESIGN

Database Design Overview:

The database is organized into logical groups covering user management, organisation approval, volunteer applications, community assistance, emergency alerts, journey safety, trusted contacts, safety reporting, safe places, rewards, and administrative monitoring.

The table structure helps separate different types of information and supports the application's modular design. Relationships between tables allow relevant records to be associated with users, assistance requests, journeys, and other platform activities, according to the implemented schema.

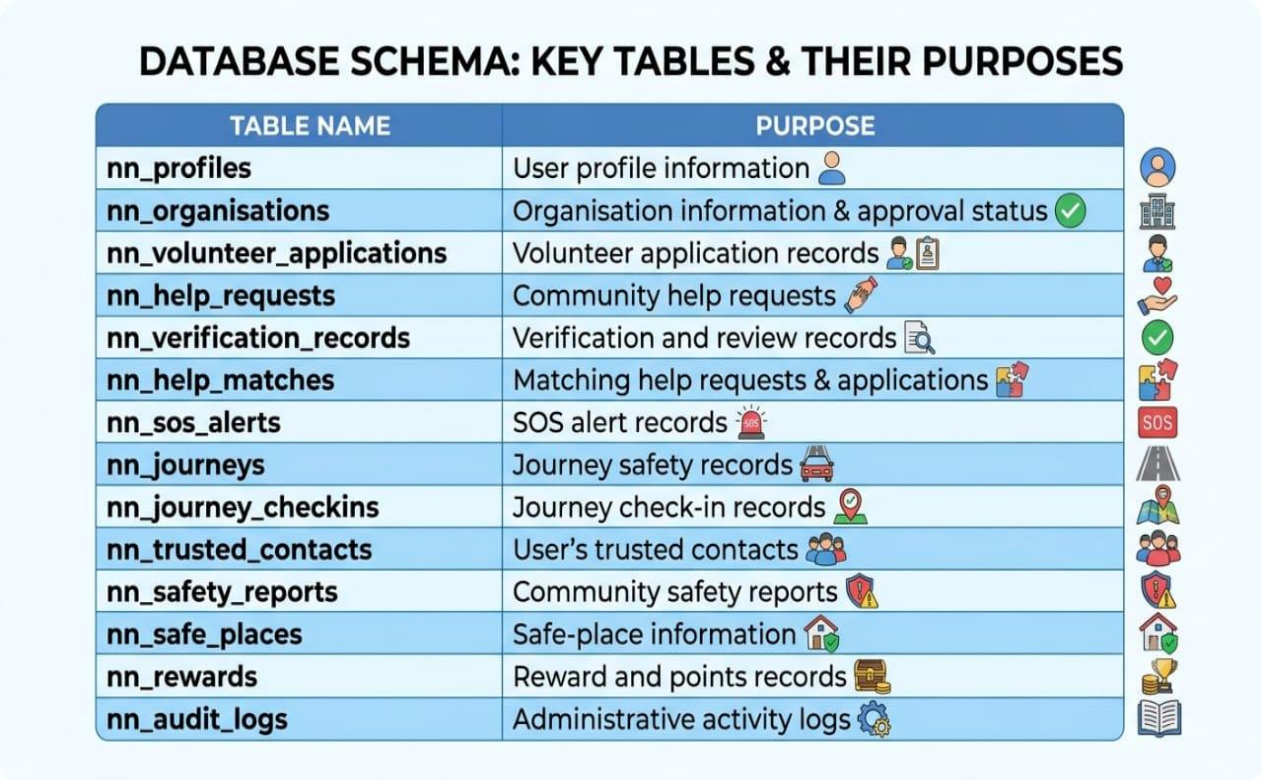
Data Security and Integrity:

Since NIRBHAYANET may process personal information and location-related records, the database should follow appropriate security and privacy practices.

These include:

- Restricting access to sensitive records.
- Applying suitable database permissions.
- Using Row-Level Security (RLS), where configured, to control access to individual records.
- Validating data before storing it.
- Protecting trusted-contact and journey information from unauthorized access.
- Maintaining administrative activity logs where implemented.
- Collecting and retaining only the information required for the application's functions.

Database table:



Conclusion:

The NIRBHAYANET database contains 14 tables that support the platform's main functional areas. Together, these tables provide a structured foundation for managing user information, community assistance, emergency-related records, journey safety, community reporting, safe places, rewards, and administrative activities.

IMPLEMENTATION DETAILS

Frontend Implementation:

- **Architecture & Framework:**

Developed using React.js with a component-based structure to ensure clean code separation, high reusability, and maintainability across the user interface.

Utilizes modern JavaScript (ES6+) and JSX for rendering dynamic, reactive UI elements and managing application views.

- **Styling & User Experience:**

Implemented Tailwind CSS for utility-first, responsive styling, ensuring the application adapts seamlessly to various screen sizes and device viewports.

Designed clean navigation layouts, interactive forms, and dashboard interfaces focusing on user accessibility and visual consistency.

- **State Management & API Communication:**

Handled component-level reactivity and lifecycle workflows using built-in React hooks (useState, useEffect).

Integrated asynchronous communication channels to connect the frontend seamlessly with the FastAPI backend services for real-time data exchange.

Backend Implementation:

The backend of NIRBHAYANET was developed using Supabase, a Backend-as-a-Service (BaaS) platform powered by PostgreSQL.

The objective was to create a structured, secure and scalable database system that could support the application's major modules, including:

- User profiles and identity management
- Organisation management
- Volunteer applications
- Community help requests
- Verification records
- SOS alert records
- Journey tracking

- Journey check-ins
- Trusted contacts
- Community safety reports
- Verified safe places
- Rewards management
- Administrative audit records

The backend was designed around separate database tables for each major functional module, with Row Level Security (RLS) to restrict access to sensitive user-owned information.

Technology Stack:

Technology	Purpose
React.js	Frontend development
JavaScript / TypeScript	Application logic
Tailwind CSS	Responsive styling
Supabase	Backend services
PostgreSQL	Database management
Supabase JavaScript Client	Frontend-backend communication
RLS and SQL	Database access control
Node.js and npm	Development environment
Git and GitHub	Version control

Supabase Project Configuration:

A dedicated Supabase project named NIRBHAYANET was created.

Backend configuration included:

1. Creating a Supabase organisation and project.
2. Configuring the PostgreSQL database.
3. Accessing the Supabase Table Editor.
4. Creating the application's database tables.
5. Configuring primary identifiers and column data types.
6. Enabling Row Level Security on all application tables.
7. Creating access policies for different user operations.
8. Configuring the frontend connection using the Supabase JavaScript client.

Database Implementation:

- **User Profile Management:**
- **nn_profiles:**

This table stores user profile information required by the application to manage user accounts and associate users with relevant platform activities.

- **nn_organisation:**

This table stores information about organisations associated with the platform, including their approval status.

- **Volunteer and Community Assistance:**
- **nn_volunteer_applications:**

This table maintains records of volunteer applications submitted to participate in the community assistance network.

- **nn_help_requests:**

This table stores requests submitted by users seeking community assistance.

- **nn_verification_records:**

This table maintains records related to verification and review activities for relevant users, applications, or organisations.

- **nn_help_matches:**

This table records the matching of help requests with suitable volunteer applications, supporting the community assistance process.

- **Emergency and Journey Safety:**
- **nn_sos_alerts:**

This table stores records associated with SOS alerts generated through the application's emergency-support functionality.

- **nn_journeys:**

This table maintains journey safety records associated with users' journeys.

- **nn_journey_checkins:**

This table stores journey check-in records that support the journey monitoring process.

- **nn_trusted_contacts:**

This table stores information about trusted contacts associated with users to support trusted-contact functionality.

- **Community Safety and Safe Places:**
- **nn_safety_reports:**

This table stores community safety reports submitted through the platform, helping maintain records of reported safety concerns.

- **nn_safe_places:**

This table stores information about safe places listed on the platform to support location-based safety information.

- **Rewards and Administrative Monitoring:**
- **nn_rewards:**

This table maintains records related to reward points and recognition activities supported by the platform.

- **nn_audit_logs**:

- This table records administrative activities to support monitoring, accountability, and review of actions performed within the system.

Security and Integration:

The React frontend communicates with Supabase through its JavaScript Client. Database operations are subject to configured permissions and RLS policies.

RLS helps restrict access to sensitive information according to the applicable policies. Proper policy configuration and testing are necessary to protect user profiles, trusted contacts, journey records, and other private data.

Summary:

NIRBHAYANET integrates React.js, Supabase, and PostgreSQL to provide a structured application architecture. Its modular database supports the platform's major safety-related features, while RLS provides a mechanism for controlling database access.

TESTING AND VALIDATION

Introduction

Testing and validation are important for checking the functionality, usability, and reliability of the NIRBHAYANET application. Testing focuses on verifying frontend components, database operations, and access-control mechanisms.

Testing Approach

The following testing methods are considered for the application:

1. Frontend Testing:

Checks page layouts, navigation, buttons, forms, and responsiveness across different screen sizes.

2. Functional Testing:

Checks the implemented features, including user profiles, help requests, journey records, safety reports, and SOS alert records.

3. Database Testing:

Verifies database operations and the storage and retrieval of records from the PostgreSQL database through Supabase.

4. Security Testing:

Checks Row Level Security (RLS) policies to ensure users can access only the data permitted by the configured rules.

5. Integration Testing:

Checks communication between the React frontend and Supabase services, including appropriate handling of errors and responses.

Validation

Validation focuses on determining whether the implemented features meet the project's requirements.

The application should be checked for correct input handling, appropriate access permissions, consistent database operations, and clear feedback to users. Safety-related features should also be tested to confirm their actual behaviour under expected conditions.

Conclusion

Testing and validation help identify functional issues, database errors, and access-control weaknesses in NIRBHAYANET. The application should undergo appropriate testing before deployment. Actual test results should be documented based on the tests performed.

RESULTS AND DISCUSSION

Results

NIRBHAYANET was developed as a community-powered women's safety platform using React.js, Supabase, and PostgreSQL. The application includes a structured database with 14 tables to support its major modules, including user profiles, volunteer applications, help requests, SOS alerts, journey safety, trusted contacts, safety reports, and safe places.

The frontend provides an interface for users to interact with the application's features, while Supabase supports database operations and access control through configured Row Level Security (RLS) policies.

Discussion

The project brings multiple women's safety features together within a single digital ecosystem. Its modular database structure supports organized data management, while the React-based frontend supports a responsive user experience.

The use of Supabase and PostgreSQL provides a foundation for maintaining application records and managing data access. However, the effectiveness of the safety features depends on correct implementation, appropriate security policies, reliable information, and sufficient testing.

Further improvements may include comprehensive security testing, usability evaluation, and validation of emergency and community-assistance workflows.

Summary

NIRBHAYANET establishes a technical foundation for an integrated women's safety platform. Further testing and validation are necessary to assess its reliability, security, usability, and readiness for real-world deployment.

ADVANTAGES AND LIMITATIONS

Advantages

1. Unified Platform: Brings multiple women's safety features together in one application.

2. Community Assistance: Supports requests for assistance from the community.

3. Journey Safety: Provides features for managing journey information and check-ins.

4. Trusted Contacts: Supports maintaining trusted-contact information.

5. Safe-Place Information: Organizes information about designated safe places.

6. Structured Database: Uses PostgreSQL to organize application data.

7. Access Control: Supabase Row Level Security (RLS) helps restrict unauthorized database access when correctly configured.

8. Scalable Foundation: The modular architecture provides a foundation for future improvements.

Limitations

1. Internet Dependency: Many application features require an internet connection.

2. Verification Challenges: Volunteer and organisation information requires appropriate verification procedures.

3. Location Accuracy: Location-based features may be affected by device settings, network conditions, and GPS accuracy.

4. Emergency Response: The application cannot guarantee immediate assistance or replace official emergency services.

5. Privacy Risks: Personal and location-related information requires strong security and careful access management.

6. Testing Requirements: Security, usability, and reliability require further testing before real-world deployment.

7. Operational Costs: Hosting, third-party APIs, and maintenance may introduce costs as the project grows.

Summary

NIRBHAYANET provides a foundation for an integrated women's safety ecosystem. However, its effectiveness depends on reliable implementation, appropriate verification, data protection, and continued testing. It is intended to complement existing emergency services rather than replace them.

FUTURE SCOPE

NIRBHAYANET can be enhanced in the future through the following improvements:

1. Mobile Application: Develop dedicated Android and iOS applications for easier access.

2. Real-Time Location Sharing: Improve journey tracking and location sharing with trusted contacts, subject to user consent.

3. Emergency Notifications: Integrate reliable notification services for timely alerts.

4. Verified Community Volunteers: Introduce stronger identity verification and volunteer training procedures.

5. Interactive Safety Map: Enhance maps with updated safe-place information and nearby assistance options.

6. AI-Based Safety Insights: Explore AI to identify patterns in reported safety concerns, with appropriate privacy safeguards.

7. Multilingual Support: Add regional languages to improve accessibility.

8. Enhanced Security: Strengthen authentication, database policies, privacy controls, and security testing.

9. Public-Service Integration: Explore partnerships with relevant organisations and emergency-service providers, subject to their approval and technical feasibility.

10. Scalability: Improve infrastructure to support more users and locations.

Conclusion

These enhancements could help NIRBHAYANET become a more accessible, reliable, and scalable women's safety platform. Future development should prioritize user privacy, verified assistance, security, and coordination with official emergency services.

CONCLUSION

NIRBHAYANET is a community-powered women's safety platform designed to bring journey safety, trusted contacts, community assistance, SOS alerts, and safe-place information together within a unified digital ecosystem.

The project uses React.js for the frontend and Supabase with PostgreSQL for backend services and data management. Row Level Security (RLS) provides a mechanism for controlling access to sensitive information.

NIRBHAYANET aims to make safety-related resources more accessible while encouraging responsible community participation. However, its effectiveness depends on proper implementation, reliable information, strong privacy protections, and thorough testing.

With further development and collaboration with relevant organisations, NIRBHAYANET has the potential to evolve into a more accessible and reliable women's safety solution that complements official emergency services.

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